Not For You Page (NFYP)

Investigating Anti-Retention Recommendation Systems for Digital Well-Being

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Background and Problem Statement

Feed-based recommendation engines drive modern platforms:

- Explicitly engineered around retention and engagement optimization
- Continuous architectures lead to digital burnout and reduced user agency

Limitations of existing interventions:

- Rely almost entirely on external interface controls (e.g., app timers)
- Recommendation algorithms themselves remain unexamined as tools for self-regulation

The "Not For You Page" (NFYP)

Core concept and objectives:

- An alternative feed model that inverts retention objectives
- Systematically reduces content resonance and introduces intentional friction
- Evaluates if recommendation logic can directly support voluntary platform exit

Research Goal:

- Bridge the gap between digital well-being interventions and underlying algorithmic logic

Related Work

Engagement-First Algorithmic Design:

- TikTok-style feeds rely on collaborative and content-based filtering to maximize retention (Zhou, 2024)
- Documented negative outcomes include compulsive use, echo chambers, and identity formation (Smith & Short, 2022; Kuncoro & Hasanah, 2024)

Limitations of Existing Well-Being Interventions:

- Current tools focus almost exclusively on interface-level friction or external timers (Haliburton et al., 2024; Almoallim & Sas, 2022)
- **The Gap:** Existing interventions leave the underlying recommendation objective entirely unexamined

System Mechanics and Experimental Setup

System Mechanics:

- Initial profiling assessment establishes baseline interest vectors
- Algorithm systematically suppresses primary interests and samples uneventful media

Experimental Design:

- Sample of 40 active daily users ($N = 40$), naive to disengagement hypothesis
- Standardized mobile test environment logging detailed interaction telemetry
- 1:1 randomized assignment to Control (standard feed) vs. Treatment (NFYP)

Primary Session Telemetry

Primary interaction metrics across experimental conditions ($N = 40$).

Metric	Control	NFYP	Test (t)	p-value
Mean Session Duration	54.2 min	38.6 min	$t(38) = 4.62$	< 0.001
Mean Dwell Time / Post	6.7 s	4.9 s	$t(38) = 4.31$	< 0.001
Total Items Viewed	478.3	359.2	$t(38) = 4.45$	< 0.001
Rapid-Skip Rate (posts/min)	3.1	4.9	$t(38) = -4.82$	< 0.001

Disengagement Dynamics and Exit Motivations

Retention and Exit Analysis:

- Kaplan-Meier survival curves show clear retention divergence
- At 30 minutes, 34% of NFYP participants exited vs. 16% in control

Self-Reported Exit Motivations:

- NFYP Group: 54% cited content irrelevance or mild disinterest
- Control Group: 64% cited external schedule constraints

Discussion and Implications

Key Takeaways:

- Algorithmic friction successfully prompts voluntary session termination
- Demonstrates feasibility of embedding well-being objectives into core recommendation logic

Limitations and Future Work:

- Evaluation limited to single-session setup with $N = 40$ sample size
- Long-term user adaptation and habituation remain to be explored
- Need for research on longitudinal adaptation and dynamic friction mechanisms

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